PalmSource™ 99

Conduit Kits for Java

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Overview

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- Synchronization Overview
- HotSync® Architecture
- HotSync Architecture and Java
- Java Conduit Development
- Java Conduit Classes
- Installing Java Conduits
- Summary

Synchronization Overview

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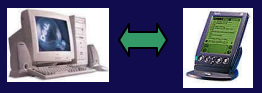
- Desktop Synchronization
  - Handheld and synchronization engine on local workstation
  - Data usually located on local workstation
- Network Synchronization
  - Handheld connected to any computer on network
  - Synchronization runs on handheld user's workstation
  - Data accessible from handheld user's workstation
- Server Synchronization
  - Handheld connected directly to network (or client)
  - Synchronization runs on servers
  - Data and synchronization engines distributed

Synchronization Dataflow

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Mirror image	Ideal for applications that run on both platforms and allow modifications on both; makes data identical on both platforms. (address book and date book applications)	Your conduit must provide conflict resolution solutions when the same record has been modified on both platforms. This must occur without user intervention.
One directional Upload / Download	For applications that run on both platforms but only allow data modification on one; copies the data to the other platform.	Requires less time, and should be used when possible.
Transaction based	For applications that need to perform additional processing on the desktop computer between record synchronizations. (e-mail, web, PQA, etc.)	Only use this when required, because it slows down the HotSync® process.
Back up	For applications that don't have PC components; simply copies the handheld data to the PC.	Can use the Backup conduit provided by Palm Computing®.

HotSync® Architecture

HotSync Manager



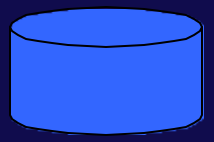
Conduit API
- OpenConduit()
- CfgConduit()

Application-Specific Conduit



Conduit Base Classes
- CSynchronizer
- CBaseMon

Data Source



- Local/networked data file
- Local/networked database (ODBC)
- Web-based data source
- E-mail or news server
- Dial-in connection

SyncManager API
- SyncOpenDb()
- SyncReadNextModified()
- SyncWriteRec()

Handheld



Contains applications/databases
- Unique creator id for each application
- Built-in HotSync device client
- HotSync process initiated by the device

HotSync® Architecture and Java

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HotSync Manager

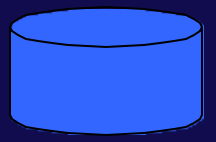


Jsync.dll

Application-Specific Conduit



Data Source



Conduit Class
- Conduit.open()
- Conduit.name()
- Conduit.configure()

Conduit Base Class
- Category
- Log
- SyncProperties
- ConfigureConduit

SyncManager Wrapper Class
- SyncManager.openDB()
- SyncManager.readNextModifiedRec()
- SyncManager.writeRec()

- Local/networked data file
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Handheld



Contains applications/databases
- Unique creator id for each application
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Jsync.dll

- Acts as an interface between native HotSync Manager calls and your Java classes
- Loads the Java Virtual Machine (JVM)
- Provides an interface from C++ Conduit API calls to Java
  - OpenConduit() → Conduit.open()
  - CfgConduit() → Conduit.configure()
  - GetConduitName() → Conduit.name()

Features of Palm OS® Conduit Development Kit /Java

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- SyncManager Class
  - Database/record access
  - Wraps SyncManager API
- Conduit Class
  - Implements base class of Conduit API
- Conduit Sample Projects
  - Source code to upload data from Palm Computing standard applications

Conduit Entry Points

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Required

- Conduit.open()
  - Synchronization entry point
  - Analogous to OpenConduit()
- Conduit.name()
  - Returns name of conduit for display purposes
  - Analogous to GetConduitName()

Optional

- Conduit.configure()
  - Display any UI necessary to configure conduit
  - Analogous to CfgConduit()

→ Note: File linking not supported

SyncManager API

Database/Record Structure

Database Name
Database Creator
Database Type
Number of Records
Last Sync Date
AppInfo Block Pointer
Sort Info Pointer
Data Record 1
Data Record 2
Data Record n

Use SyncManager Class to access remote databases and records.

SyncManager.readNextModifiedRec()

RawRecordInfo Class

m_RecId
m_RecIndex
m_attribs
m_CatId
m_pBytes



Unique Id	Category	Attribs	Raw Record Data

Java Conduit Development

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- Create your record class by modifying the MemoRecord class (extending AbstractRecord)
- Create a RecordManager Class by modifying the RecordManager class from the MemoCond or AddrCond samples
- CategoryManager can be used unchanged in most cases
- Create a Conduit class that implements your conduit
- Add a Sync Action Configuration dialog to your conduit if necessary

Native Synchronization Objects

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- Conduit Class
  - Implements the conduit class including conduit entry points

Native Synchronization Objects

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  - Implements the conduit class including conduit entry points
- Manager Class
  - MyManager is the main synchronization logic
  - Reads and writes handheld data
  - Example is RecordManager

Native Synchronization Objects

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- Conduit Class
  - Implements the conduit class including conduit entry points
- Manager Class
  - MyManager is the main synchronization logic
  - Reads and writes handheld data
  - Example is RecordManager
- Writer Class
  - Reads and writes data to the desktop data source

Native Synchronization Objects

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- CategoryManager
  - Synchronizes category information
  - Does not usually need to be modified

Native Synchronization Objects

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- CategoryManager
  - Synchronizes category information
  - Does not usually need to be modified
- ConfigureConduit
  - Conduit configuration class
  - Implements any UI necessary

Conduit Control Flow

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- HotSync® Manager calls your Conduit.open() function
  - Examine SyncProperties class for sync information
- Your Conduit.open() function instantiates a Manager class object and performs the following tasks:
  - Register the conduit
  - Create the Writer object
  - Synchronize categories information using CategoryManager
  - Synchronize application information (AppInfo block)
  - Synchronize records using Fast Sync, Slow Sync, HH overwrites desktop, Desktop overwrites HH, or No Action
  - Log results to HotSync log using Log class

Conduit Development Steps

Creating a Java Conduit from Palm® Samples

1. Define your record as MyRecord class extending AbstractRecord

```
public class MemoRecord extends AbstractRecord {

    String memo;

public String getMemo() {
        return memo;
    }
}
```

Conduit Development Steps

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Creating a Java Conduit from Palm® Samples

2. Modify sample RecordManager code to synchronize your records

```
public RecordManager(SyncProperties props, int db, Class recordClass){
        this.props = props;
        this.db = db;
        this.recordClass = recordClass;
        hhRecords = new Vector();
        slowSync = false;
    }

 public Vector fastSyncData(Vector pcRecords, Vector archivedRecords,
Vector backupRecords) throws SyncException, IOException
{
}
```

Conduit Development Steps

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Creating a Java Conduit from Palm® Samples

3. Modify sample Writer class to read/write your data on desktop

```
public class MemoWriter {…}
```

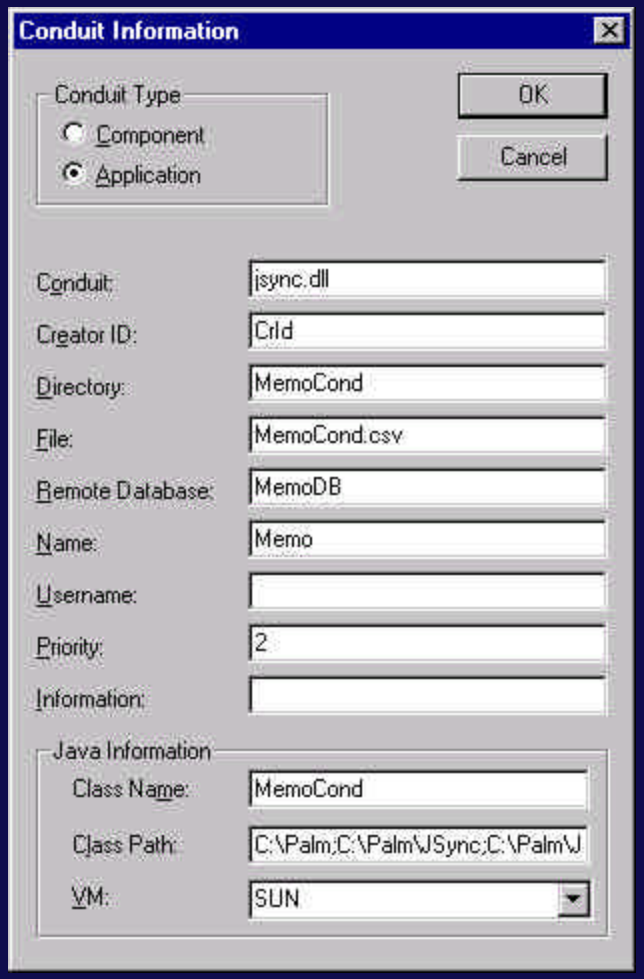
4. Modify sample ConfigureConduit class to present configuration dialog

```
public class ConduitConfigure implements WindowListener, ActionListener
{…}
```

Java Conduit Registration

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- Jsync.dll is used for all Java conduits
- The ‘Name’ field (required) is used to display your conduit in the ‘Custom’ dialog
- ‘Class Name’ is the name of your class implementing ‘Conduit’ and must be found in ‘Class Path’
- Set desired JVM



Java Conduit Development Hints

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- Use the documentation for the Conduit Development Kit for Windows in addition to the CDK/J documentation
- Visit the Knowledge Base for sample code and FAQs (http://www.palm.com/devzone)
- Source level debugging of Java conduits not possible
- RecordManager and CategoryManager classes provide a good starting point for core synchronization logic class in Java

Installing Java Conduits

- Use Palm-supplied APIs for conduit installation
  - Install Aide API
  - Conduit Manager API
  - User Manager API (UserData.dll)

  Note: These APIs are included in the Conduit Development Kit (CDK 3.0) for Windows
- A JSync installer is provided in the CDK/J package
  - The Jsync package (jsync.dll and jsync.zip) must be installed by your Java conduit installer—it is not standard on the Palm Desktop software
- See sample install code on Knowledge Base

Installation/Management of Java Conduits

- Install Aide API (CDK 3.0 for Windows)
  - Permits easy installation/management of Palm Computing Platform applications
- Conduit Manager API (CDK 3.0 for Windows)
  - Permits easy installation/management of conduits
- User Manager API (CDK 3.0 for Windows)
  - Permits easy management of Palm users
- HotSync API (HSAPI) (CDK 3.0 for Windows)
  - Programmatic control of HotSync process

Summary

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- The Palm Computing® Platform is the leader in handheld computing and information management
- Synchronization (Data Exchange) plays an important role in handheld computing (personal and enterprise information management)
- The HotSync® architecture is adaptable to many synchronization scenarios and includes support for Java conduits on Windows
- The Palm Computing CDKs provide a complete set of tools for conduit development for Windows


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